

2025년 1학기 데이터통신 11주차

과제 보고서

-로또 통신-



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# 과제 해결

```
Windows PowerShell
PS F:\학교\4학년\1학기\데이터통신\과제\11th> python .\server.py
Listening on <socket.socket fd=416, family=AddressFamily.AF_INET, type=SocketKind.SOCK_DGRAM, p
proto=0, laddr=('127.0.0.1', 3034)>
Received from ('127.0.0.1', 53160)
Numbers in bag: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45]
Send [5, 8, 13, 20, 31, 35] to ('127.0.0.1', 53160)
Received from ('127.0.0.1', 53160)
Numbers in bag: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45]
Send [20, 24, 25, 27, 39, 40] to ('127.0.0.1', 53160)
Received 1 5 10 14 from ('127.0.0.1', 53160)
Numbers in bag: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45]
Send [1, 5, 10, 14, 29, 42] to ('127.0.0.1', 53160)
Received 42 13 4 6 17 9 from ('127.0.0.1', 53160)
Numbers in bag: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45]
Send [4, 6, 9, 13, 17, 42] to ('127.0.0.1', 53160)
Received 1 31 18 from ('127.0.0.1', 53160)
Numbers in bag: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45]
Send [1, 5, 6, 7, 18, 31] to ('127.0.0.1', 53160)
```

```
Windows PowerShell
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새로운 기능 및 개선 사항에 대 한 최신 PowerShell을 설치 하세요! https://aka.ms/PSWindows

PS F:\학교\4학년\1학기\데이터통신\과제\11th> python .\client.py
Ready to send using <socket.socket fd=428, family=AddressFamily.AF_INET, type=SocketKind.
SOCK_DGRAM, proto=0>
Data:
Send to (127.0.0.1,3034)
Received 5 8 13 20 31 35 from ('127.0.0.1', 3034)
Data:
Send to (127.0.0.1,3034)
Received 20 24 25 27 39 40 from ('127.0.0.1', 3034)
Data: 1 5 10 14
Send 1 5 10 14 to (127.0.0.1,3034)
Received 1 5 10 14 29 42 from ('127.0.0.1', 3034)
Data: 42 13 4 6 17 9
Send 42 13 4 6 17 9 to (127.0.0.1,3034)
Received 4 6 9 13 17 42 from ('127.0.0.1', 3034)
Data: 1 31 18
Send 1 31 18 to (127.0.0.1,3034)
Received 1 5 6 7 18 31 from ('127.0.0.1', 3034)
Data: |
```

## Server

```
print(f'Listening on {sock}')
while True:
    bag = []
    for i in range(1, 46):
        bag.append(i)
    data, client=sock.recvfrom(2**16)
    data=data.decode('utf-8')
    data = data.strip()
    print(f'Received {data} from {client}')
    print(f'Numbers in bag: {bag}')

    if data: # 데이터가 비어있지 않을 때만 처리
        dataNum = data.split(' ')
        length = len(dataNum)
        if DEBUG:
            print(f'Numbers in data: {dataNum}')
            print(f'Length of data: {length}')
        for i in range(length):
            bag.remove(int(dataNum[i]))
        output = random.sample(bag, 6-length)
    else: # 데이터가 비어있으면 6개 전부 랜덤 선택
        length = 0
        output = random.sample(bag, 6)

    dataNum = list(map(int, data.split(' '))) if data else []
    final = dataNum + output
    final = sorted(final)

    final_str = ' '.join(map(str, final))
    sock.sendto(final_str.encode('utf-8'), client)
    print(f'Send {final} to {client}')
```

서버에 대한 코드이다. 클라이언트로부터 데이터를 받으면 로또 번호가 있는 가방에서 클라이언트가 보낸 번호의 수만큼을 6에서 빼고 나머지에 해당할 수를 랜덤 추출하여 선택한다. 고른 번호들은 다시 클라이언트에 전송해준다.

## Client

```
1  import socket
2  FLAGS = _ = None
3  DEBUG = False
4  def main():
5      if DEBUG:
6          print(f'Parsed arguments {FLAGS}')
7          print(f'Unparsed arguments {_}')
8      sock=socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
9      print(f'Ready to send using {sock}')
10     while True:
11         data = input('Data: ').strip()
12         sock.sendto(data.encode('utf-8'),(FLAGS.address,FLAGS.port))
13         print(f'Send {data} to ({FLAGS.address},{FLAGS.port})')
14         data, server = sock.recvfrom(2**16)
15         data = data.decode('utf-8')
16         print(f'Received {data} from {server}')
17
18     if __name__ == '__main__':
19         import argparse
20         parser=argparse.ArgumentParser()
21         parser.add_argument('--debug',action='store_true',
22             help='The present debug message')
23         parser.add_argument('--address',type=str,default='127.0.0.1',
24             help='The address to serve service')
25         parser.add_argument('--port',type=int,default=3034,
26             help='The port to serve service')
27         FLAGS,_=parser.parse_known_args()
28         DEBUG=FLAGS.debug
29         main()
```

클라이언트에 대한 코드이다. 실습 코드를 활용하였다. 서버에 보낼 데이터를 입력하면 고를 숫자를 제외한 만큼 서버에서 랜덤하게 나머지 번호들을 골라주면 클라이언트는 해당 번호를 받는다.